

FIELD STUDY PREPARED FOR CRW ENGINEERING GROUP, INC.

CRW's BD and PM staff hand-assemble every state, federal, and tribal proposal from scratch while the firm is running six simultaneous engineering openings, and a human-reviewed AI draft-and-checklist tool could return scarce PE and PM hours to billable work.

CRW Engineering Group runs on a steady stream of public-sector, tribal, and rural-infrastructure task orders, the kind of work that lives and dies on RFP and SOQ responses. At the same time, CRW's own careers page keeps an evergreen 'always hiring' posture and, as of this study, had six paid openings live at once across CAD, inspection, and three separate PE disciplines, at an 80-person firm. The people who would normally write the past-performance, staffing, and qualifications sections of a proposal are the same people the firm is short on. Nobody among CRW's three closest verified competitors (PND Engineers, R&M Consultants, DOWL) shows any public sign of AI-assisted business-development tooling either, so the field around CRW is as undifferentiated on this as CRW is.

Other

Anchorage and Palmer, Alaska

2026-07-19

Researched and drafted by Alaska AI's agent team, reviewed by a human before it reached you

WHAT WE SAW

The homework

THE BUSINESS

CRW Engineering Group is a full-service, Alaska-based civil engineering and land surveying firm with more than 80 staff across offices in Anchorage, Palmer, and Olympia, Washington. It offers one-stop planning, permitting, surveying, engineering, and construction-management services, spanning airport, civil, structural, electrical, mechanical, environmental, and geotechnical engineering, water and wastewater, transportation, and rural infrastructure. Its client base is heavily public-sector and tribal, municipalities, Alaska Native villages, and federal agencies.

THE REAL PAIN

CRW is always on the lookout for exceptional engineers, surveyors, technicians, and support staff... Or contact us with your resume if you don't currently see an open position relevant to your skillset.

That evergreen careers-page posture sat alongside six simultaneously posted paid openings: CAD Technician, Electrical Engineer PE, Construction Inspector, Civil Engineer EIT Transportation, Civil Engineer PE Water/Wastewater, and Civil Engineer PE Transportation. For an 80-person firm, six open roles spanning CAD, inspection, and three separate PE disciplines at once points to a firm running short-handed against its project load broadly, not in one team.

Source: <https://www.crweng.com/careers/> and <https://www.indeed.com/cmp/Crw-Engineering-Group>

THE FIELD AROUND THEM

- **PND Engineers, Inc.** none found
[source](#)
- **R&M Consultants, Inc.** A GIS StoryMap for the Alaska Native Heritage Center, data visualization, not AI. No AI, chatbot, or automation otherwise found. [source](#)

AI IN THIS INDUSTRY NOW

- Bechtel runs an internal AI tool queried over its own project documents, generating punch lists and pulling priorities from thousands of pages, a working production tool, not a pilot.
[source](#)

- **DOWL** none found [source](#)

- 63 percent of ACEC member firms report having an AI strategy in place or actively being developed as of Q1 2025, an 11-point jump over the prior year. [source](#)

- Only 27 percent of surveyed AEC professionals report actually using AI in their operations, most firms with any AI activity are stuck between intent and real use. [source](#)

THE OPPORTUNITY

When a state, federal, or tribal RFP or SOQ solicitation drops, assemble a compliant, well-written response fast, pulling from CRW's own past project descriptions, resumes and qualifications, and boilerplate.

This scored highest of the opportunities the research supported. It touches every discipline's pipeline in a firm whose revenue model is fee-for-service task orders, and none of CRW's three verified competitors show any public AI-assisted business-development tooling. It beats the more loudly-cited recruiting pain on fit, that pain is real but is a conventional ATS problem, not a genuine AI opportunity.

WHAT THEY USE TODAY

Manual, template-based assembly from prior proposals and resume files, done by the same PE and PM staff the six open roles show are already stretched.

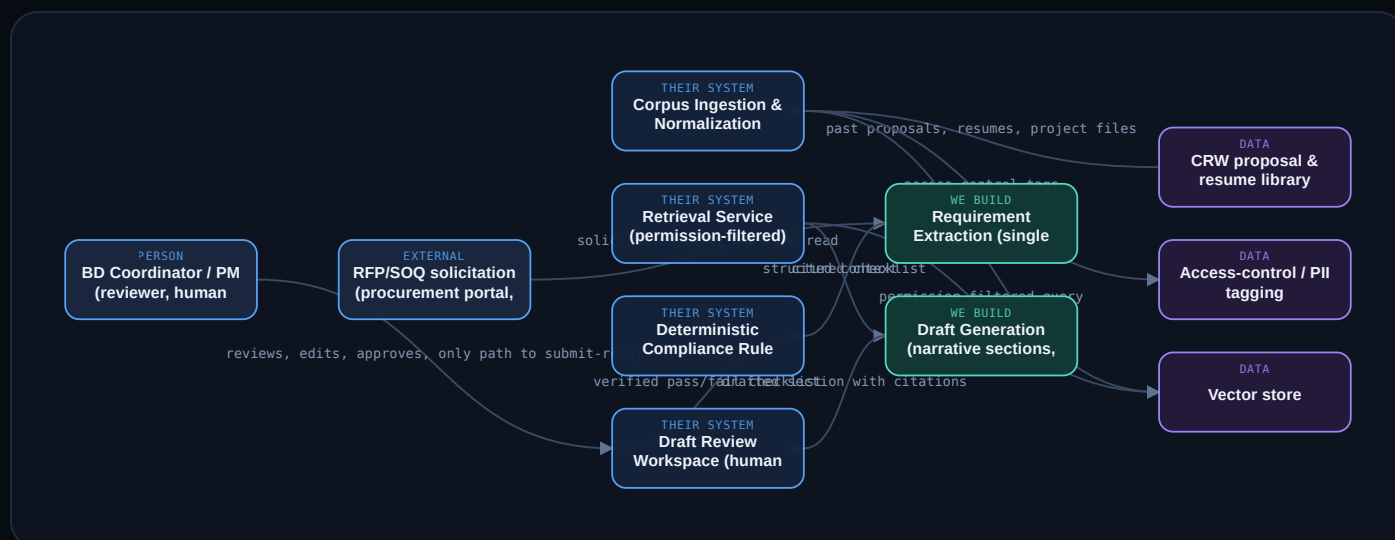
WHY A CHANGE IS REALISTIC

The push is real, scarce PE and PM staff spend hours hand-assembling formulaic sections under public-sector deadline pressure. The pull is a first draft pulled from CRW's own past-performance library. The honest friction is that public solicitations carry strict compliance formatting, which raises real quality-anxiety about AI-drafted content, addressed below by keeping compliance checks deterministic rather than AI-judged.

THE RECOMMENDED BUILD

Proposal & SOQ Engine

A BD coordinator or PM feeds in a new RFP or SOQ. A single AI read of the solicitation extracts the objective compliance requirements, page limits, required forms, mandatory sections, into a checklist. A deterministic rule engine, not a second AI pass, verifies that checklist. Separately, a retrieval-plus-drafting step pulls from CRW's own ingested proposal library, resumes, and project descriptions to draft first-pass narrative sections, past performance, staffing, qualifications, each sentence traceable back to its source document. A human reviews, edits, and is the only one who can move a draft to submitted. No automated submission path exists or is planned.



■ Person ■ External ■ Their system ■ We build ■ Data

The solicitation enters on the left through a deterministic compliance gate. CRW's own proposal and resume library feeds a permission-aware retrieval layer that only the draft generator reads from. Every draft lands in a review workspace where a human, and only a human, can approve it.

WHERE AI FITS, AND WHERE IT DOES NOT

This is a workflow-tier build, not an autonomous agent, retrieval plus one drafting call, with a mandatory human gate before the one costly action. It earns that tier because matching a solicitation's specific requirement language against CRW's own past-performance content is a real retrieval-and-drafting problem a static template can't do as well. Where we

BUILD VS BUY

Buy the commodity, the embedding model, vector store, base language models, authentication, and observability tooling are all hosted services. Build the differentiator, the corpus ingestion pipeline that normalizes CRW's actual proposal and resume files, the permission-filtered retrieval layer, the deterministic compliance rule engine, and

would NOT use AI, the compliance checklist itself, page limits, required forms, mandatory sections, is an objective yes-or-no check. It runs as a deterministic rule pass after one AI read extracts the requirements, never as a second round of AI judgment. The same discipline ruled out two other candidates entirely, recruiting and applicant screening is a conventional ATS problem where a false negative costs a labor-scarce firm capacity it cannot spare, and a rural-client status portal rests on no sourced client complaint and would sit oddly against Alaska's relationship-based client culture. Neither needs AI, and neither is part of this build.

the review workspace that enforces the human gate.

THE PLAN

What we would build, scoped

CRW runs on a steady diet of state, federal, and tribal RFPs and SOQs. Every response needs past-performance write-ups, staffing bios, and qualifications narratives assembled from scratch, plus a compliance checklist gotten right. CRW is visibly running lean, six paid openings were live at once across CAD, inspection, and three PE disciplines at an 80-person firm, meaning the senior staff who write these sections are also the staff CRW is short on. This is inferred from CRW's revenue model and staffing pattern, not a proposal-specific complaint heard directly from CRW, and this study says so plainly.

GOALS

- Cut the hands-on time PM and BD staff spend hand-assembling narrative proposal sections.
- Catch objective compliance requirements before the human reviewer's first read, not after.
- Keep full human review and edit authority before submission intact.
- Stand up a permissioned, structured proposal and resume corpus CRW owns.

NON-GOALS, ON PURPOSE

- Any path to automated or AI-assisted submission of a proposal. (A wrong claim or missed compliance item can disqualify a bid or create false-claims exposure. A human reviews and submits every time.)
- AI judgment calls on whether a proposal is compliance-complete. (Objective yes/no checks run as a deterministic rule pass, never a second round of AI judgment.)
- Any drafting or review of stamped engineering content. (That work carries the sealing engineer's personal, non-delegable liability. This build never touches it.)
- Recruiting or applicant-screening automation. (A conventional ATS problem, not a genuine AI fit, evaluated and set aside.)
- A client-facing project-status portal for rural community clients. (Rests on

weaker evidence and, if built at all, should be a plain dashboard, not a conversational layer.)

HOW WE WOULD MEASURE SUCCESS

METRIC	BASELINE	TARGET	BY WHEN
PM and BD staff hours spent hand-assembling narrative proposal sections per response	Assumed, to be established with a 4-6 week time-log before Phase 1 launch	At least a 30 percent reduction versus the logged baseline	First full quarter of proposal cycles after launch
Objective compliance misses caught only at final review or after submission	Assumed zero formal tracking today	Zero missed mandatory items past the deterministic checklist stage	First two full proposal cycles after launch
Share of first-pass AI-drafted sections rated a usable starting point by reviewers	Zero, no AI draft exists today	At least 70 percent rated usable-as-a-starting-point	First quarter of use
Pursuits submitted per quarter at the same staffing level	Unknown, to be pulled from CRW's own BD tracking	A measurable lift without added headcount	Two quarters following launch

PHASE ONE, MUST HAVE

- Ingest and permission CRW's existing proposal library, resumes, and project descriptions into a structured, access-controlled corpus.
- A single AI read of an incoming solicitation extracts objective compliance requirements into a checklist.
- A deterministic rule pass verifies that

DELIBERATELY LATER

- An internal search assistant over CRW's project archive and regulatory library, riding the same ingestion pipeline.
- Extending drafting coverage to more solicitation types.
- A won or lost-bid feedback loop.

checklist.

- Retrieval-plus-drafting of first-pass narrative sections, tied back to source documents.
- A mandatory human review-and-edit gate before submission.
- A CRW-signed-off acceptance bar for what counts as a usable first draft.

WHAT WE WOULD NEED FROM YOU

- Access to CRW's existing proposal library and past submissions.
- Staff resumes and qualifications records, and a clear owner of resume PII permissioning.
- A named BD coordinator or PM as day-to-day point of contact.
- Four to six weeks of proposal-cycle hours logged for a real baseline.
- A recent sample of solicitation documents to validate the compliance-checklist logic.
- Sign-off that a human review-and-submit gate is the permanent operating model.

RISKS, AND HOW WE WOULD HANDLE THEM

RISK	MITIGATION
The proposal-assembly pain is inferred, not a directly sourced CRW complaint.	Validate with a short baseline time-log before committing full Phase 1 budget.
CRW's proposal library and resumes are likely scattered and inconsistent.	Corpus ingestion is its own budgeted first phase, validated with a small pilot first.
A wrong or missed compliance requirement can disqualify a bid.	Mandatory human sign-off before submission, compliance checks run as deterministic rules.

RISK

MITIGATION

If reviewers find drafts unusable, the tool adds friction instead of removing it.

Define the acceptance bar with CRW staff up front, measure first-draft-usable rate from day one, halt further phases if it misses the bar.

OPEN QUESTIONS WE WOULD RESOLVE TOGETHER

- What is CRW's actual PM and BD hours spent per proposal cycle today? (CRW BD lead, via a short time-log)
- How clean and centralized is CRW's existing proposal and resume library? (CRW IT or BD coordinator)
- What counts as a first draft good enough to route to a human for review? (CRW BD lead, defined jointly before the build starts)

THE NUMBERS

Honest ROI, not a hero number

WHAT IT COSTS TO RUN, FIVE YEAR VIEW

The real ask is the Phase-1 pilot, not the full five-year build. The pilot is the spike plus walking skeleton, a compliance checklist and drafted narrative sections on one live pursuit, using a small hand-loaded set of past proposals and resumes rather than the full library. Assumed pilot cost, a quarter to a third of the full Year-1 implementation estimate below, roughly \$20,000 to \$30,000. The full build is priced only as context for what a staged commitment could grow into, and its later phases are gated on the pilot's measured results, not funded up front. Full five-year total cost of ownership, not sticker price, implementation and integration in Year 1 (the biggest line, since CRW's proposal and resume corpus is unconfirmed-clean) runs an assumed \$65,000 to \$95,000, training and change management adds roughly \$7,000 to \$18,000, ongoing run cost (inference, monitoring, evaluation, human review, corpus upkeep) runs an assumed \$18,000 to \$30,000 a year, and a 15 to 20 percent contingency sits on top. Modeled 5-year TCO ranges roughly \$176,000 (lean) to \$235,000 (most likely) to \$300,000 (conservative, if the corpus is messier than assumed).

WHERE THE VALUE COMES FROM

- **Senior PE and PM proposal-drafting hours saved per pursuit** CAPACITY Converts to cash by redeploying freed PE and PM time back onto billable stamped work, CRW is running six simultaneous openings, short-staffed rather than positioned to defer a hire.
- **BD coordinator compliance-assembly hours saved per pursuit** CAPACITY Converts to cash via redeploy of the coordinator's freed hours to additional pursuit prep.
- **Reduced risk of a compliance-technicality bid disqualification** CASH Booked as avoided cost, the smallest and least-certain of the three quantified lines. Modeled assuming 1 to 2 near-miss or lost pursuits a year at an assumed \$8,000 to \$15,000 in sunk pursuit cost each, with the checklist assumed to cut that risk 50 to 70 percent. No CRW-specific disqualification history exists, this is a stated assumption, not a measured figure.

CONSERVATIVE

Full run-rate benefit around \$15,000 a year, reached only after a partial first year (roughly 25 to 35 percent of run-rate while implementation and corpus ingestion ramp up).

Cumulative benefit over five years is about \$70,000 against the \$300,000 TCO, about 23 percent recovered. Does not pay back within five years.

MOST LIKELY

Full run-rate benefit around \$43,000 a year, same partial-first-year ramp.

Cumulative benefit over five years is about \$198,000 against the \$235,000 TCO, about 84 percent recovered, trending toward full breakeven around year six or seven, beyond this study's five-year window.

AGGRESSIVE

Full run-rate benefit around \$90,000 a year against the lean end of cost. Crosses breakeven at roughly 18 to 24 months and finishes five years at about \$419,000 cumulative benefit against \$176,000 cumulative cost.

Payback 18 to 24 months under aggressive assumptions on the full build. The most-likely case recovers about 84 percent of its cost over five years without fully paying back, and the conservative case recovers only about 23 percent. We are saying this plainly rather than picking the number that clears approval. This is exactly why the ask here is the small pilot above, not the full build, so CRW pays a few tens of thousands to find out which scenario its real numbers land closest to before any larger commitment.

The honest base rate

Most AI projects never move the bottom line, MIT puts it at roughly 95 percent, RAND at roughly 80 percent. This build is designed to sit in the smaller share that does, it is retrieval and one drafting call, not a chain of autonomous actions, it fits BD and PM staff's existing draft-then-review habit instead of requiring new behavior, and it keeps a human gate on the one costly action. Even with those guardrails, our own full-build numbers do not clear the conservative bar, which is exactly the honesty this method is built to surface rather than paper over. That is why the pilot, priced and scoped on its own, is the actual proposal, and further phases are funded only once the actual numbers, not the assumed ones, prove it out.

Who owns the number Brian Looney, President and the study's addressee, is the most plausible value owner and budget sponsor, since this study did not find a named BD director at CRW. The variance check, log actual PE, PM,

and BD-coordinator hours per pursuit against the baseline above for four full quarters after the pilot launches, and report actual versus modeled before funding any phase beyond it.

THE ROADMAP

By confidence, not by calendar

NOW	NEXT	LATER
Ingest and access-permission CRW's own past proposals, resumes, and project descriptions, including a PII redaction pass. Percent of the library ingested and access-controlled	Expand corpus coverage and drafting to CRW's full range of recurring solicitation types. Pursuits submitted per quarter	An internal search assistant over CRW's project archive and regulatory library, riding the same ingestion pipeline. Time for an engineer to find a prior design solution or regulation
Deterministic compliance-checklist extraction for one live RFP or SOQ. Zero missed mandatory items on the pilot pursuit	Build a won or lost-bid feedback loop. Win rate on submitted pursuits, held or improved	A first-pass regulatory and permitting checklist generator for rural water, sewer, and DOT projects. PE hours per project spent on manual regulatory review
First-pass narrative drafting for one live pursuit, with mandatory human review before anything is submitted. PE/PM hours per proposal, baseline versus after, on the pilot pursuit		

Next is earned only when Now's actuals on the pilot pursuit clear the bar, a measured reduction in hours, zero missed compliance items, and reviewers confirming the drafts were a real starting point. Later is earned only when Next's actuals clear its own bar. Two candidates the research turned up, recruiting automation and a rural client status portal, were evaluated and set aside rather than deferred, one is a conventional ATS problem and the other rests on no sourced client complaint, neither belongs on this roadmap.

The honest part, what could make this not work

- The core proposal-assembly pain is inferred from CRW's revenue model and hiring pattern, not a complaint we heard directly from CRW. We are treating it as a hypothesis to validate with a short time-log, not a confirmed fact.
- The honest ROI case only clears the payback bar under aggressive assumptions. The conservative and most-likely cases do not fully pay back within five years on the numbers we modeled. That is why we recommend starting as a small, cheaply-measured pilot rather than a full commitment.
- CRW's real proposal and resume corpus has not been inspected. If it is messier or more scattered than assumed, ingestion could cost more and take longer than the ranges above.
- No CRW competitor shows public AI-assisted business-development tooling, but the absence of a public sign is not proof CRW would be first, only that nothing was found.

WHAT THIS IS

This is the free version of how we work

If this is useful, reply and we will walk through the assumptions and the pilot scope together. No pitch, no call required first.

The full picture lives at alaskaaihq.com/services.

SOURCES

1. CRW offers start-to-finish, one-stop solutions for planning, permitting, surveying, engineering, and construction-management, more than 80 staff members, offices in Anchorage, Palmer, and Olympia WA <https://www.crweng.com/>
2. CRW's evergreen careers posture, always accepting resumes <https://www.crweng.com/careers/>
3. Six simultaneous paid openings across CAD, inspection, and three PE disciplines <https://www.indeed.com/cmp/Crw-Engineering-Group>
4. CRW leadership, Brian Looney (President), Shane Blanchard, Pete Bellezza, and the info@crweng.com contact <https://www.crweng.com/about/leadership/>
5. Competitor PND Engineers, Anchorage-headquartered civil/structural, arctic and waterfront projects, no AI found <https://www.pndengineers.com/>
6. Competitor R&M Consultants, Anchorage/Wasilla/Fairbanks/Juneau offices, GIS StoryMap for the Alaska Native Heritage Center, no AI found otherwise <https://www.rmconsult.com/>
7. Competitor DOWL, Juneau office, no AI found <https://www.dowl.com/>
8. Bechtel's internal AI tool over project documents, a production tool generating punch lists <https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/12/18/architecture-engineering-construction-sector-slow-to-adapt-ai-survey-shows>
9. 63 percent of ACEC member firms have or are developing an AI strategy as of Q1 2025, an 11-point increase year over year https://www.acec.org/wp-content/uploads/2025/10/AI-Adoption-in-AEC-Summary_Primer_October_2025.pdf
10. Only 27 percent of surveyed AEC professionals report actually using AI in their operations <https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/12/18/architecture-engineering-construction-sector-slow-to-adapt-ai-survey-shows>

Alaska AI, in-state and building. This study was prepared for CRW Engineering Group, Inc. and is not for redistribution.